Innovation & Entrepreneurship for Engineers

Dr Shankar

IIT Madras

2026

Innovation & Entrepreneurship for Engineers

Module 1
Engineers as Innovators

#1. What is Innovation ?
#2. Innovation through S & T

Module 2
Problem Discovery

#3. Finding Problems worth solving
#4. Customer Discovery & Design Thinking

Module 3
Creativity & Innovation

#5.Innovative Thinking
#6. Enhancing Innovation – AI - Collaboration

Module 4
Deep-Tech Entrepreneurship

#7. Deep – Tech Ventures
#8. Exponential & Systems Thinking

Module 5
Building a Startup

#9. Economics of Innovation & Entrepreneurship
#10. Business Models
#11. Intellectual Property

Module 6
Prototyping & Presentation

#12. Rapid Prototyping
#13. Pitch Your Idea
#14. Demo Day

Innovation & Entrepreneurship for Engineers

Innovative Thinking

Dr Shankar

IIT Madras

19 September, 2026

Innovative Thinking

- Innovation FLOW
- Discovery Questions
- Innovation FLOW – Opportunity Canvas
- CREATE
- Innovation FLOW – Idea Canvas

!nnovation FLOW

Dr Shankar

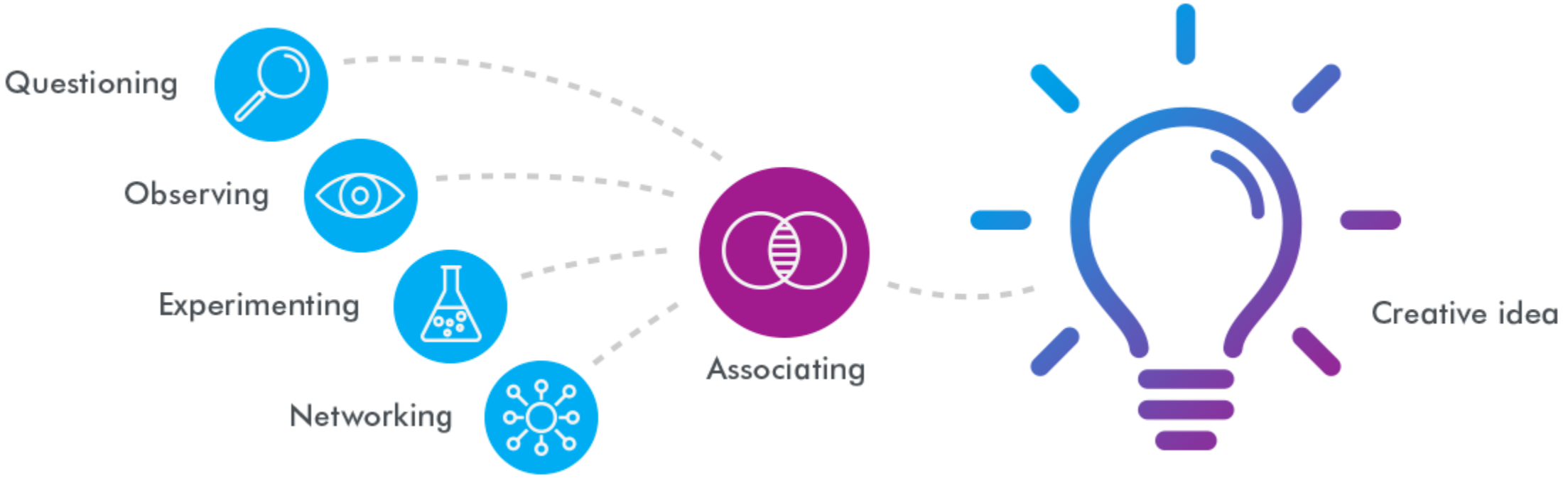
IIT Madras

September 2026

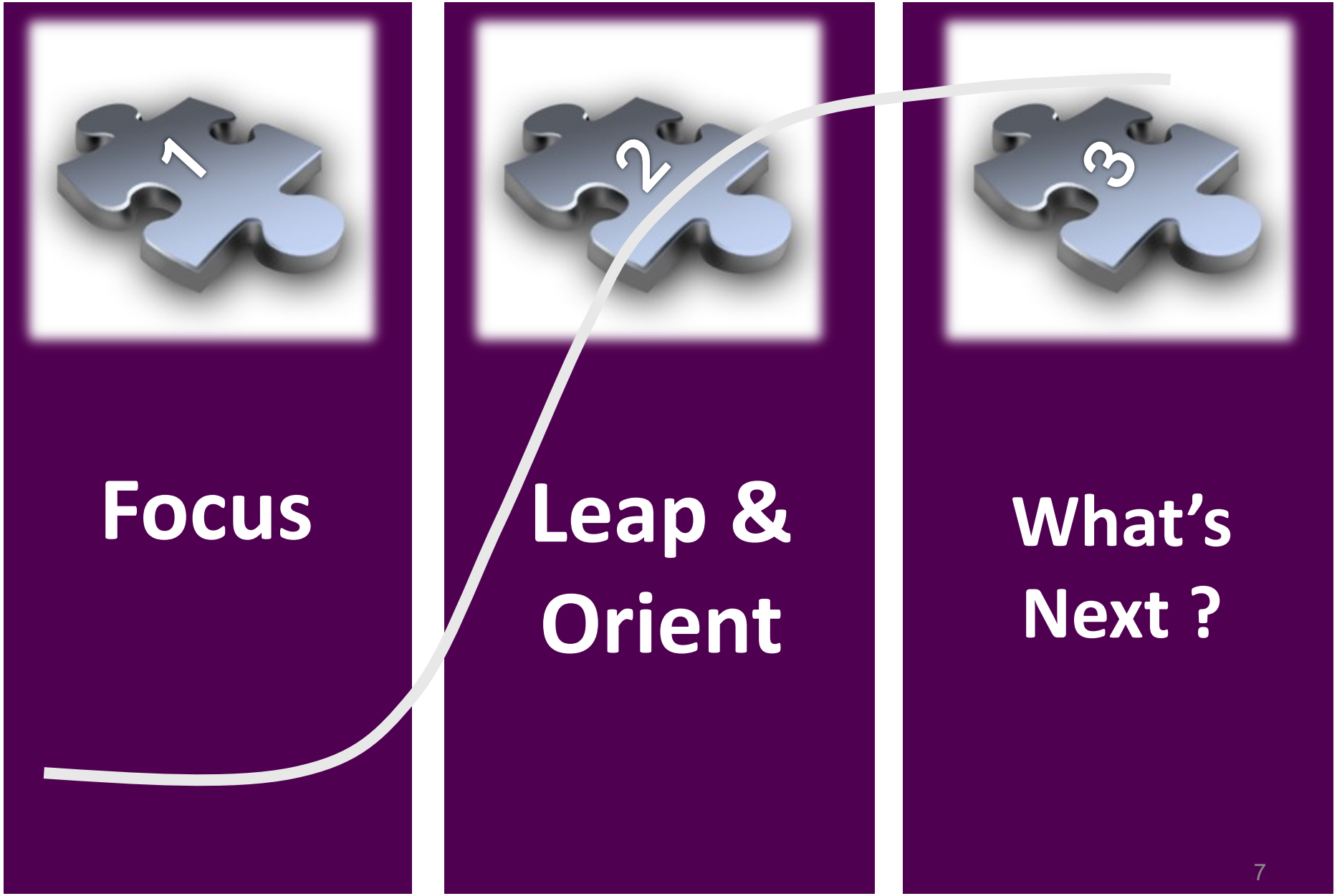


Innovator's DNA – 5 Skills

Behavioral Skills + Cognitive Skill → Disruptive Idea

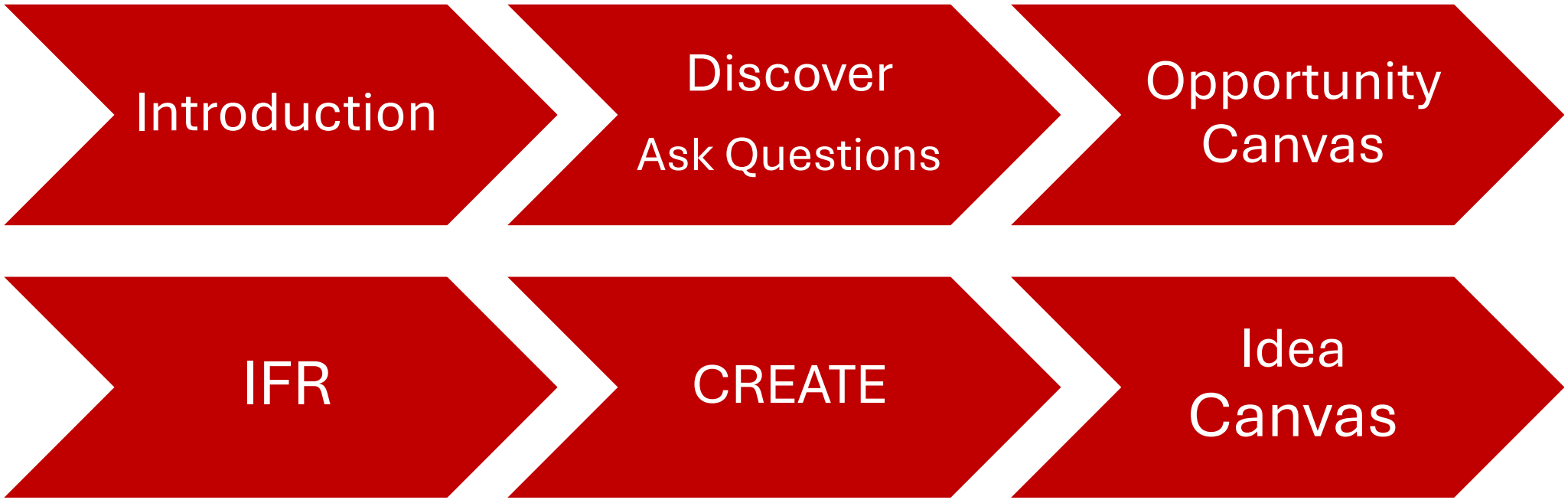


Source - The Innovator's DNA, Harvard Business Review



7

Innovation Flow – Workshop



Innovation Flow – Workshop



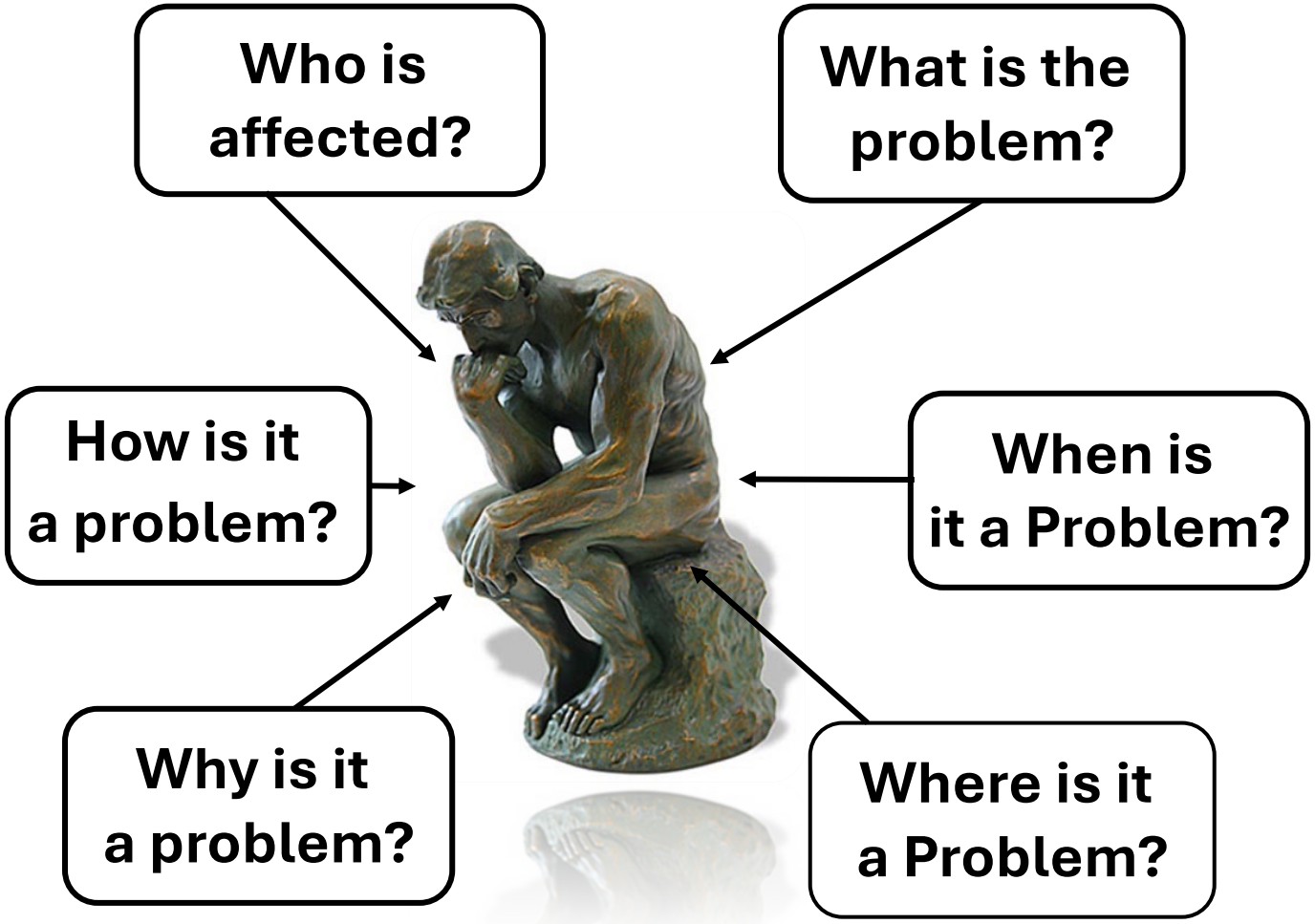
Questioning

I keep six honest serving-men
(They taught me all I knew);
Their names are What and Why and When
And How and Where and Who.

~Rudyard Kipling, The Elephant's Child



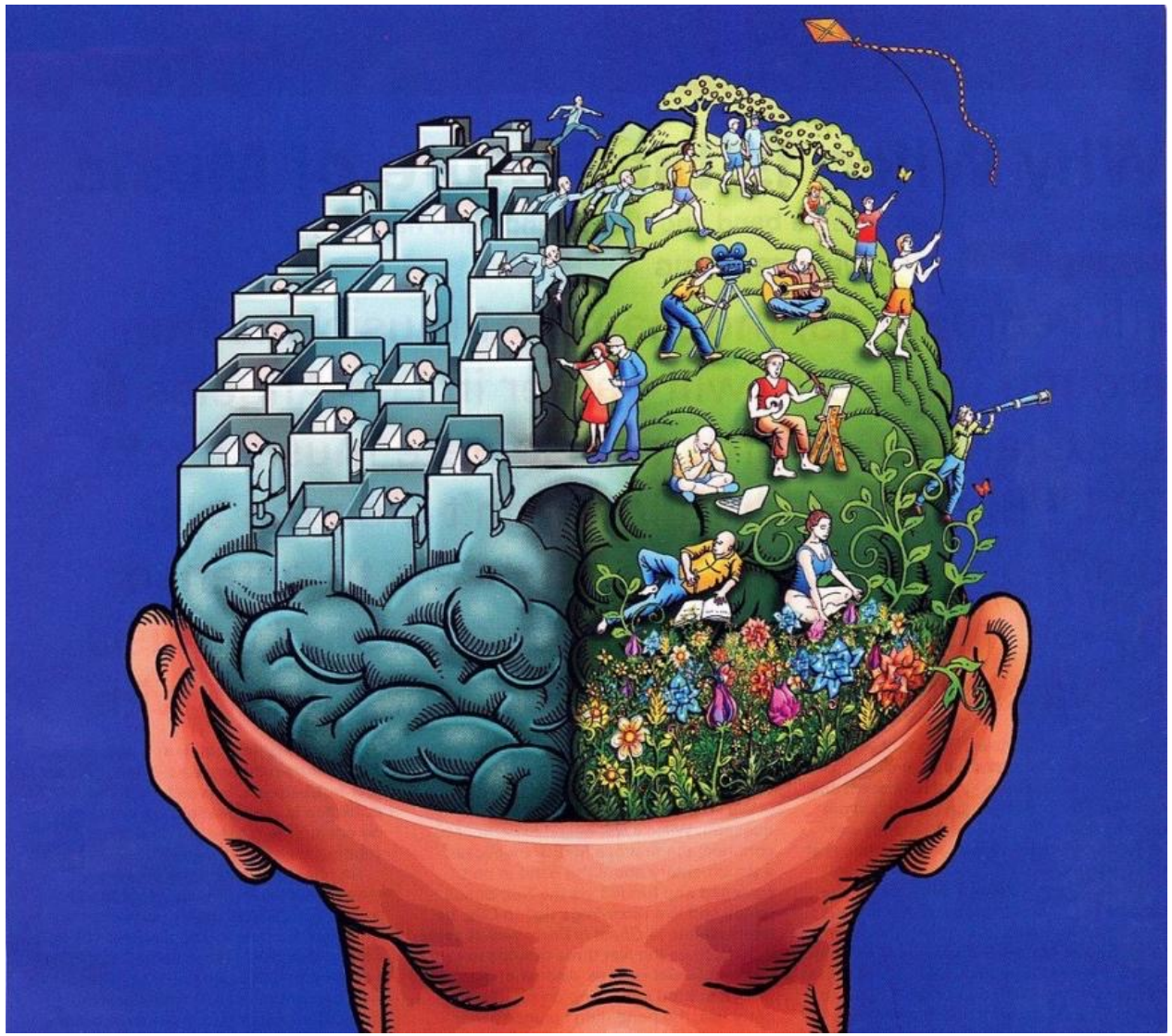
Discovery Question #1



!nnovation Flow – Opportunity Canvas

The Innovation Plot

- **Who is affected ?**
- **What is the problem ?**
- **When is it a problem ?**
- **Where is it a problem ?**
- **How is it a problem ?**
- **Why did the problem arise ?**



Zap your Logical Brain



Discovery Question #2



50 what-if questions to reimagine the future

https://www.boardofinnovation.com/staff_picks/50-what-if-questions-to-reimagine-the-future/

Inbox (3,389) - shankar.venugopa | (99+) Feed | LinkedIn | Potentialife | Nike By You. Nike IN

https://www.nike.com/in/w/nike-by-you-6ealh

Men Women Kids Customise Sale SNKRS

Search

Nike By You (55)

Hide Filters Sort By

Shoes

Gender

- Men
- Women
- Unisex

Colour

Purple Black Red
Orange Blue White
Brown Green Yellow
Multi-Colour Grey Pink



Customise

Nike Mercurial Superfly 8 Elite By You

Custom Football Boot

6 Colours

₹23,995



What if your customers could customize every single detail of your product?

The manufacturing industry is becoming increasingly capable of creating small batches (or even single units) of personalized products at very competitive prices. Imagine ways to let your customers customize their favorite product or service, and to produce/deliver it cheaply. Nike iD online store allows users to customize the look and feel of their shoes, from texture, to shape, to color, for (only) 150€. Good morning mass customization.

PotentiaLIfe - Hap....pdf

Show all

Type here to search

30°C Partly cloudy ENG 22:40 20-03-2022

Inbox (3,389) - shankar.venugopa | (99+) Feed | LinkedIn | ntialife | Nike By You. Nike IN | Duolingo - The world's best way

https://www.duolingo.com/characters

LEARN | CHARAC | SHOP | MORE

HIRAGAN

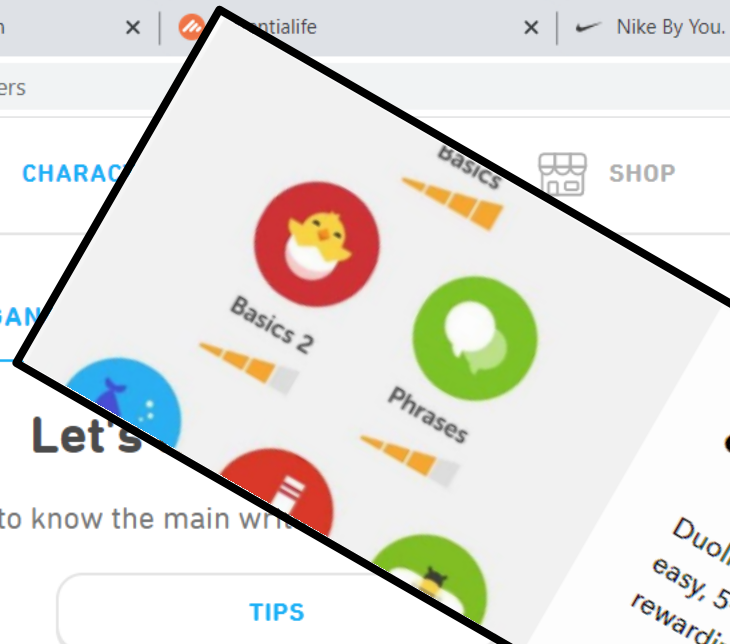
Let's

Get to know the main wri

TIPS

LEARN THE CHARACTERS

あ a	い i	う u	え e	お o
か ka	き ki	く ku	け ke	こ ko
さ sa	し shi	す su	せ se	そ so
た ta	ち chi	つ tsu	て te	と to



What if you could reframe your service into bite-size components?

Duolingo turns a lengthy, painful process (learning a new language) into a playful list of little, easy, 5-min lessons. How can you reframe a long, boring process into a list of quick, fun, rewarding activities, that the user can tackle at different times in a day?

rds!

art

Next Achievement VIEW ALL

PotentiaLIfe - Hap....pdf

Show all

Type here to search

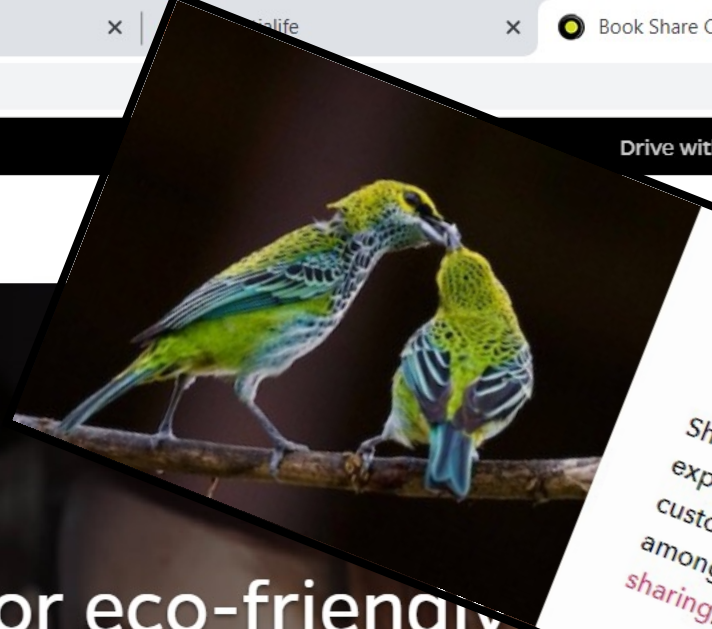
30°C Partly cloudy ENG 22:49 20-03-2022

Inbox (3,389) - shankar.venugop | (99+) Feed | LinkedIn | Book Share Cabs | Buy Share Pas

https://www.olacabs.com/share

OLA SHARE

Express | Flat-fare Zones | Share Pass | Book Now



What if your customers could share your products and services?

Sharing economy is eating the world: we share more and more assets, especially very expensive assets that we use for short spans of time. Therefore, try to imagine a setup that lets customers not own an asset, but access it only when needed, helping them splitting the cost amongst peers. Some examples: car sharing, bike sharing, scooter sharing, ride sharing, office sharing, house sharing.

Ola Share for eco-friendly at pocket-friendly rates

Car-pool with Ola Share for minimum deviations and lower AC cab fares

PotentiaLIfe - Hap....pdf

Show all

Type here to search

30°C Partly cloudy ENG 22:51 20-03-2022

Inbox (3,389) - shankar.venugopa | (99+) Feed | LinkedIn | Potentialife | Online Experiences - Airbnb

https://www.airbnb.co.in/s/experiences/online

airbnb

Become a host

Plan a trip with help from local Hosts around

Show (14)



★ 4.98 (124) · United States

Plan The Perfect New York City Vacation With Ben

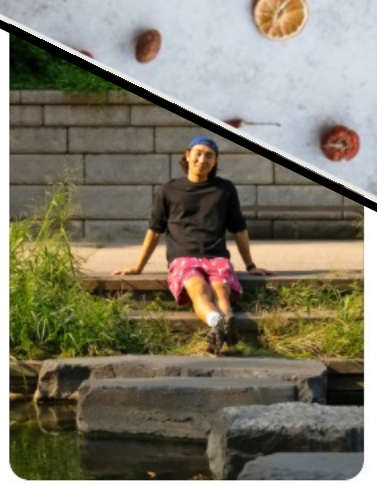
From ₹1,516/person



★ 5.0 (14) · Spain

Design your trip to Barcelona with Gemma

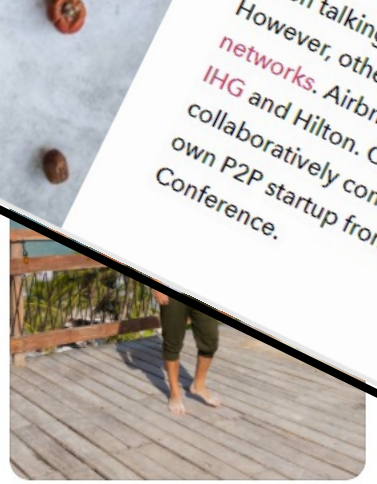
From ₹3,183/person



★ 5.0 (21) · South Korea

Craft a custom trip to Korea with Jay

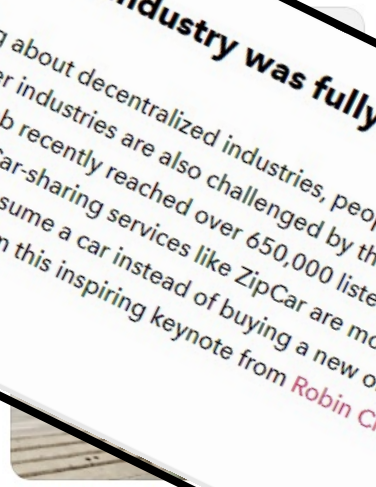
From ₹2,003/person



★ 4.67 (3) · Mexico

Explore Tulum like a local

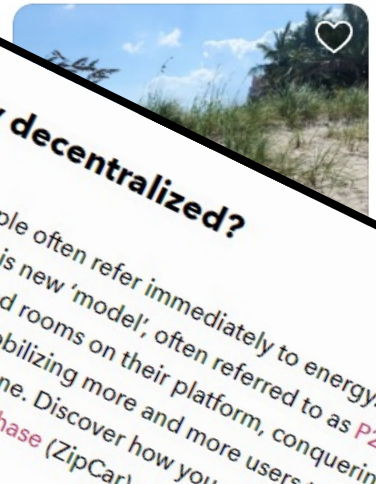
From ₹1,638/person



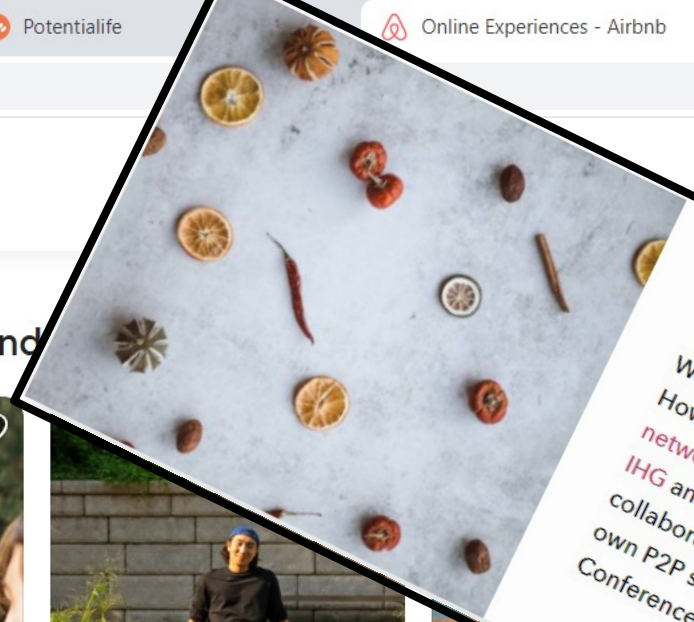
★ 4.98 (60) · Mexico

Create a custom Mexico City trip

From ~~₹1,675~~ **₹1,340**/person



From ₹



What if your industry was fully decentralized?

When talking about decentralized industries, people often refer immediately to energy. However, other industries are also challenged by this new 'model', often referred to as P2P networks. Airbnb recently reached over 650,000 listed rooms on their platform, conquering IHG and Hilton. Car-sharing services like ZipCar are mobilizing more and more users to collaboratively consume a car instead of buying a new one. Discover how you can start your own P2P startup from this inspiring keynote from Robin Chase (ZipCar) at the Lean Startup Conference.

Top sellers

Show (271)

PotentiaLIfe - Hap....pdf

Show all

Type here to search

30°C Partly cloudy

ENG

22:56

20-03-2022

https://www.haier.com/in/

Haier

Products | Buy | Service & Support | About Haier | Haier Smart Solutions



Household | Business

Refrigerators | Washing Machines | TVs | Air Conditioners | Water Heaters

Microwave Oven | Smart Home | Kitchen Appliance



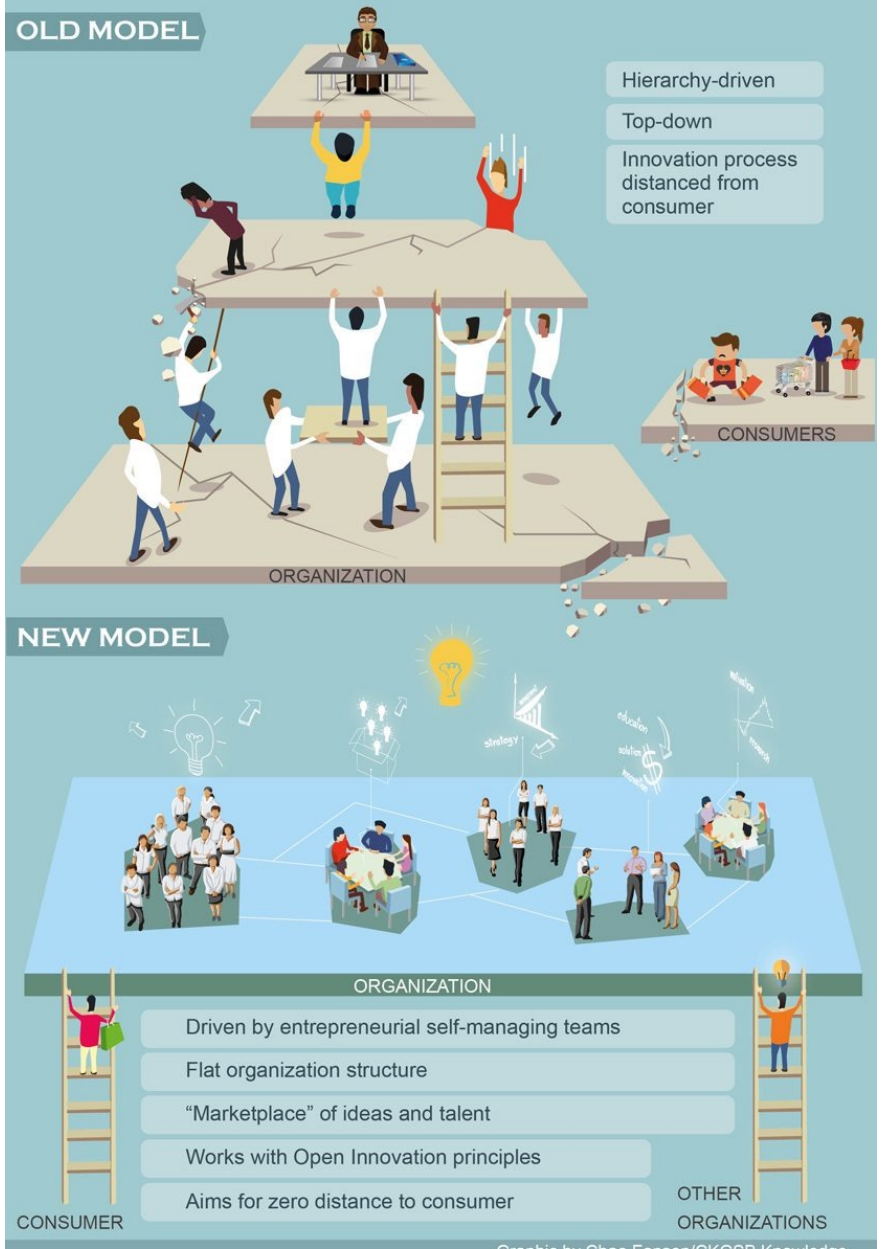


What if you launched your own competitor?

Sparks & Honey, a New York trend-spotting firm, has a wall in its office where staff post imaginative next-generation jobs. One of the new jobs people will have in 2025 is as a 'Corporate Disorganizer': big companies want to be more like start-ups, seeing innovation as vital to future profits. Young says they'll want 'corporate disorganizers' who can introduce a little "organized chaos. The disruptor will be tapping into the new systems of the collaborative economy, creating greater fragmentation and a more distributed ecosystem."

RE-ENGINEERING HAIER...

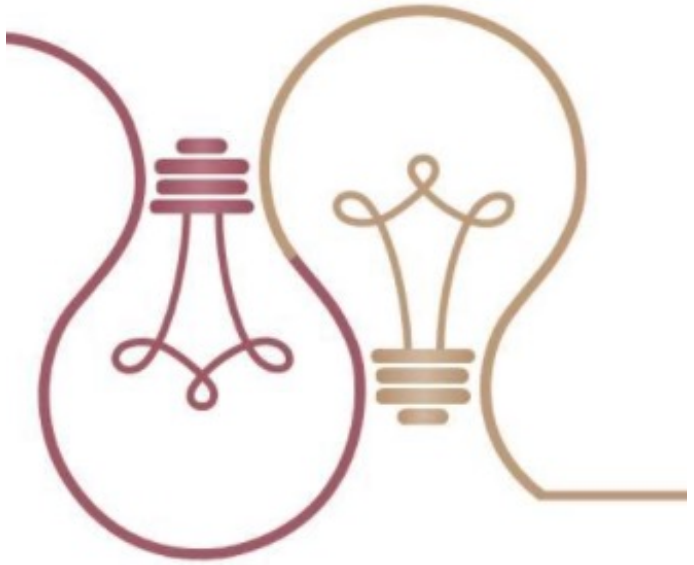
...to make it more nimble, innovative and responsive



Discovery Question #3

WIN – What if its not there ?

Reverse Thinking



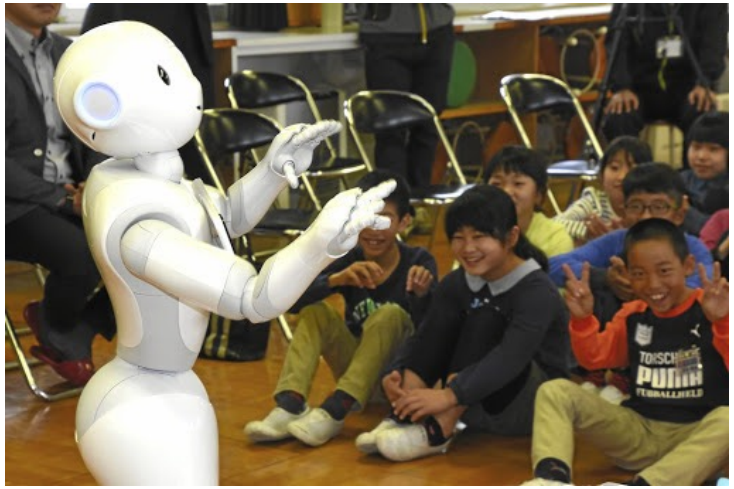
WIN - What If its Not there ?



School	WIN School
• Teacher	• No Teacher
• Building	• No Building
• Exams	• No Exams
• Students	• No Students







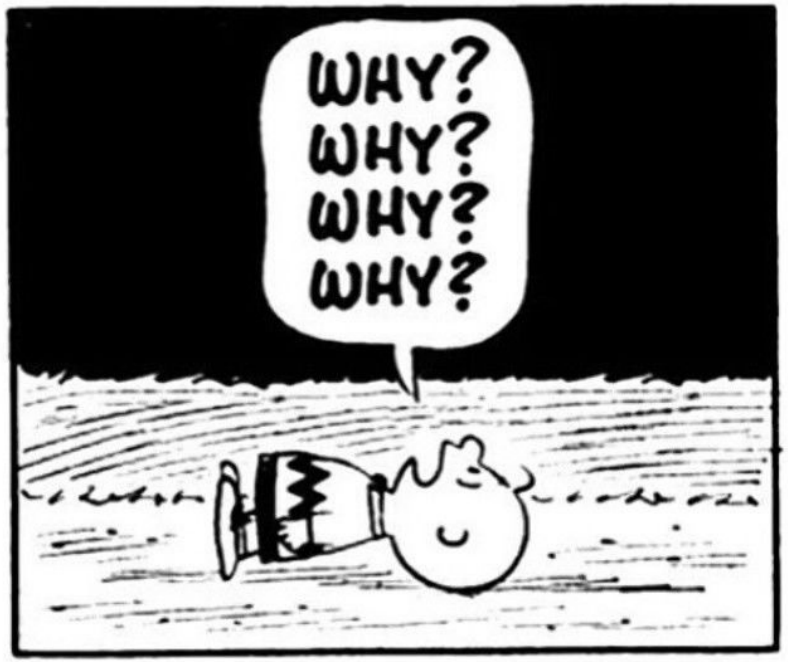




Discovery Question #4

5 Whys

Paradigm Shift



The Kingdom was lost
because
a nail was missing



1. Why was the Kingdom lost?

2. Why was the battle lost ?

3. Why could not the soldier fight well ?

4. Why was the Horse lost ?

5. Why did the horse lose its shoe ?

!nnovation Flow – Opportunity Canvas

The Innovation Plot

- **Who is affected ?**
- **What is the problem ?**
- **When is it a problem ?**
- **Where is it a problem ?**
- **How is it a problem ?**
- **Why did the problem arise ?**

Why

Why

Why

Why

Why

Innovation Flow – Workshop



!nnovation Flow – Opportunity Canvas

The Innovation Plot

- **Who is affected ?**
- **What is the problem ?**
- **When is it a problem ?**
- **Where is it a problem ?**
- **How is it a problem ?**
- **Why did the problem arise ?**

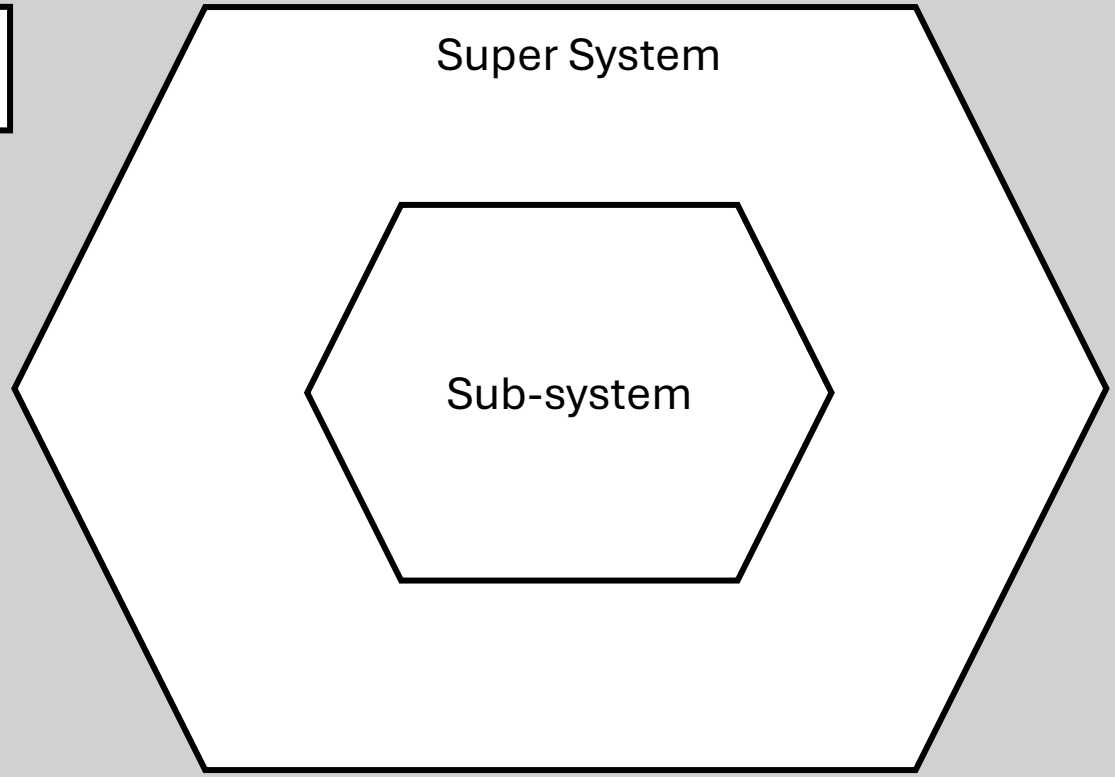
Why

Why

Why

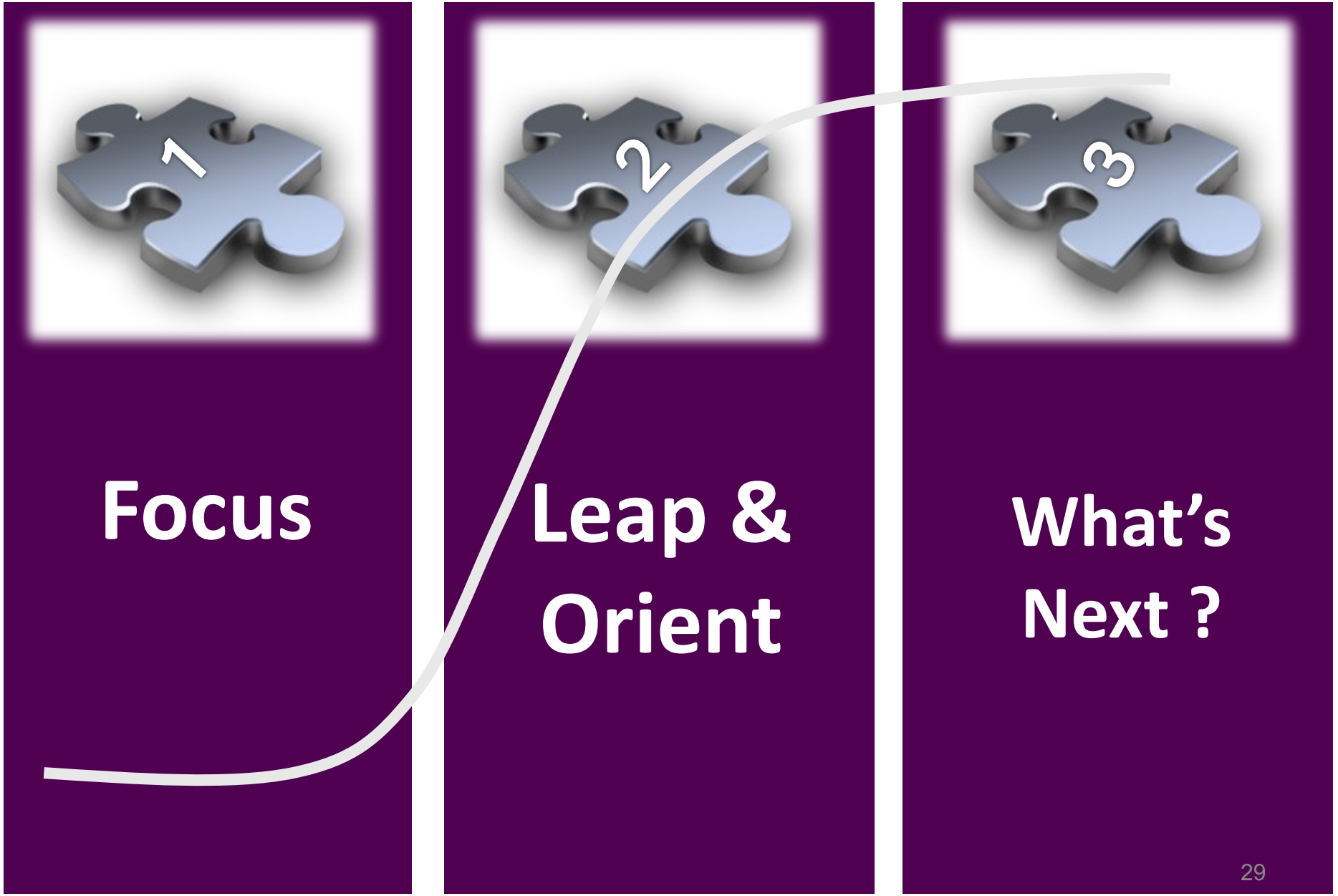
Why

Why



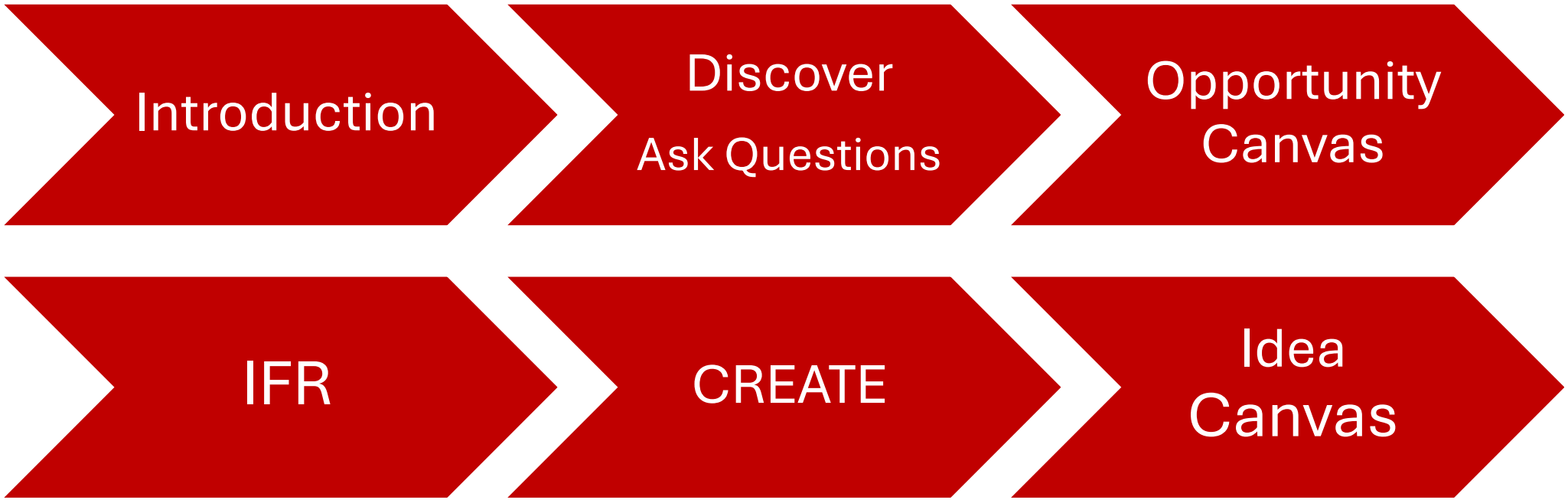
Innovative Thinking

- Innovation FLOW
- Discovery Questions
- Innovation FLOW – Opportunity Canvas
- CREATE
- Innovation FLOW – Idea Canvas



29

Innovation Flow – Workshop



Innovation Flow – Workshop



Visualize the Ideal Solution

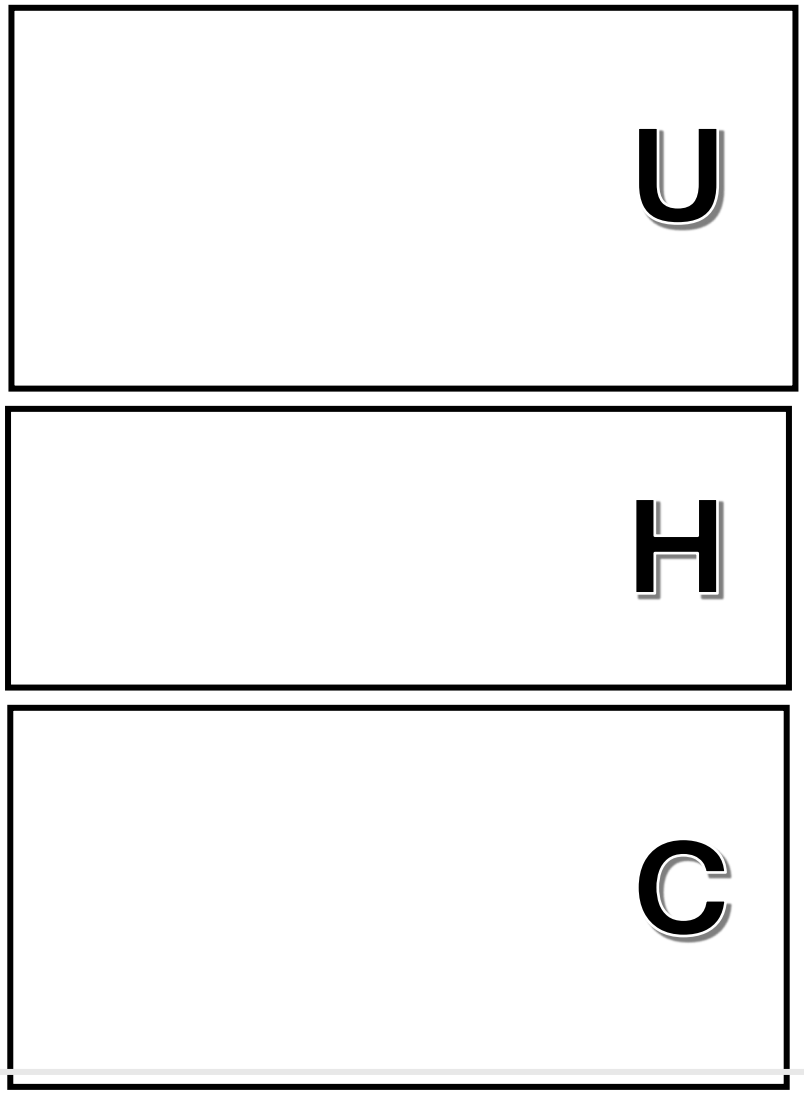
$$Ideality = \frac{\sum Useful\ effects}{\sum Harmful\ effects + Cost}$$

U – Useful features that we want to improve

H – Harmful features that we want to reduce or eliminate

C – Costs that we want to cut

IFR – Ideal Final Result (TRIZ)



!nnovation Flow - Idea Canvas

Ideal Solution :

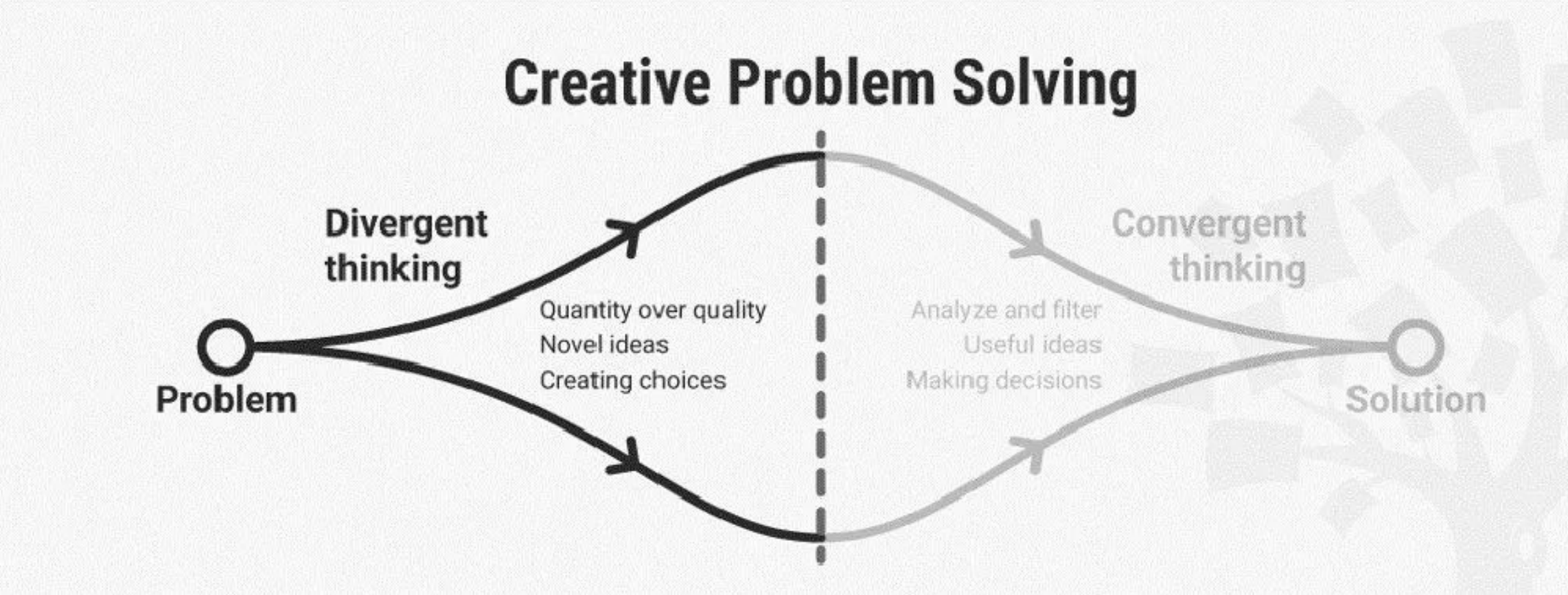
Useful functions ↑ -

Harmful functions ↓ -

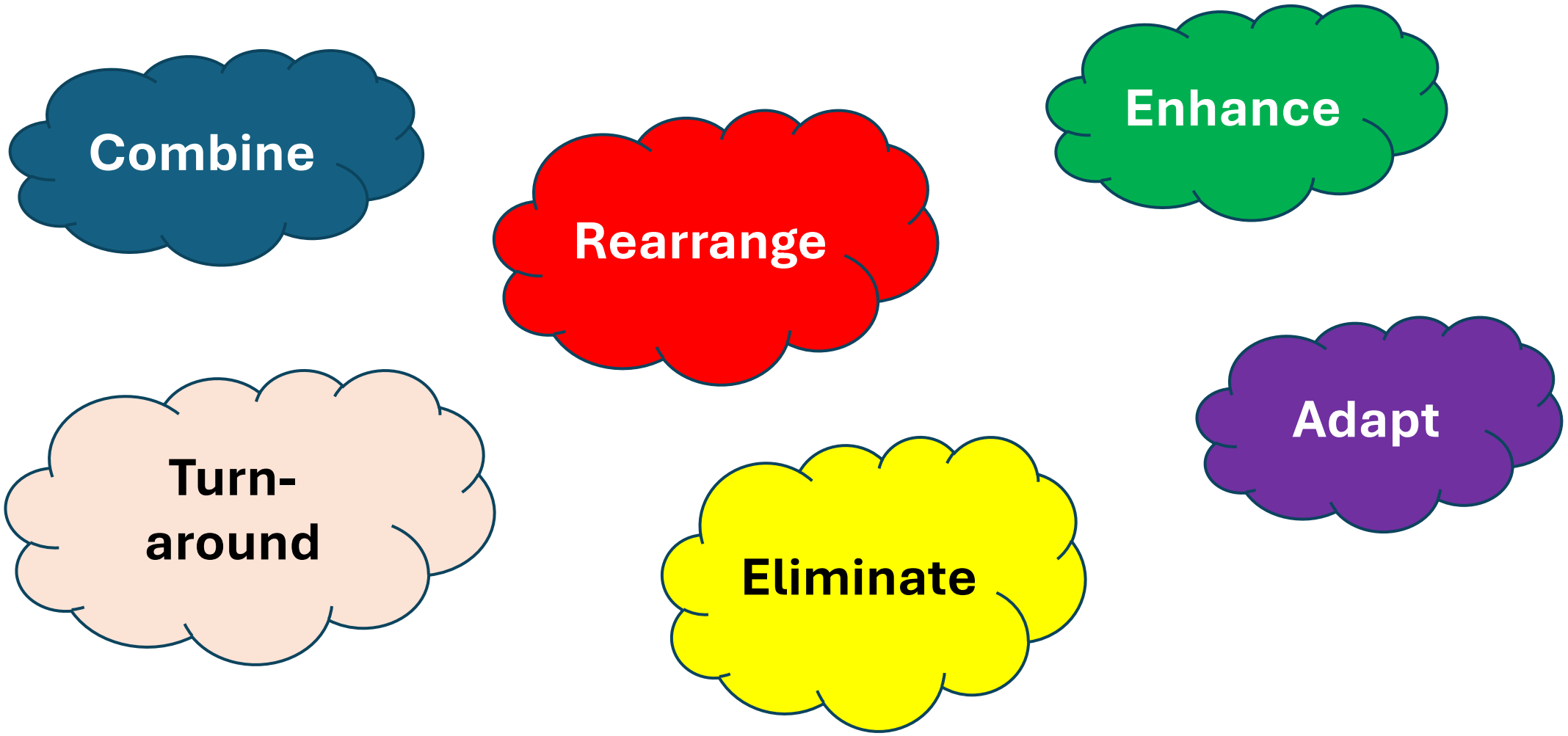
Cost ↓ -

Innovation Flow – Workshop




Creative Problem Solving
Divergent thinking
Quantity over quality
Novel ideas
Creating choices
Convergent thinking
Analyze and filter
Useful ideas
Making decisions
Problem
Solution

CREATE



CREATE - Divergent Thinking Tool

Idea Triggers	How to Improve the Useful Functions - ...	How to Reduce the Harmful Functions -	How to Reduce the Cost of .....
Combine			
Rearrange			
Enhance			
Adapt			
Turn-around			
Eliminate			

C R E A T E



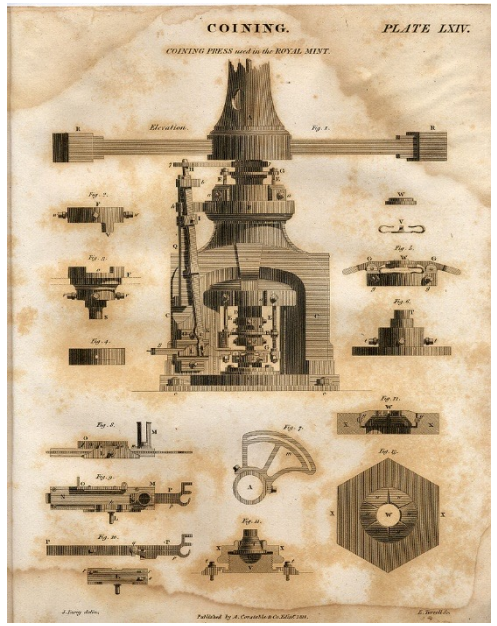
What is common to these three innovative products ?





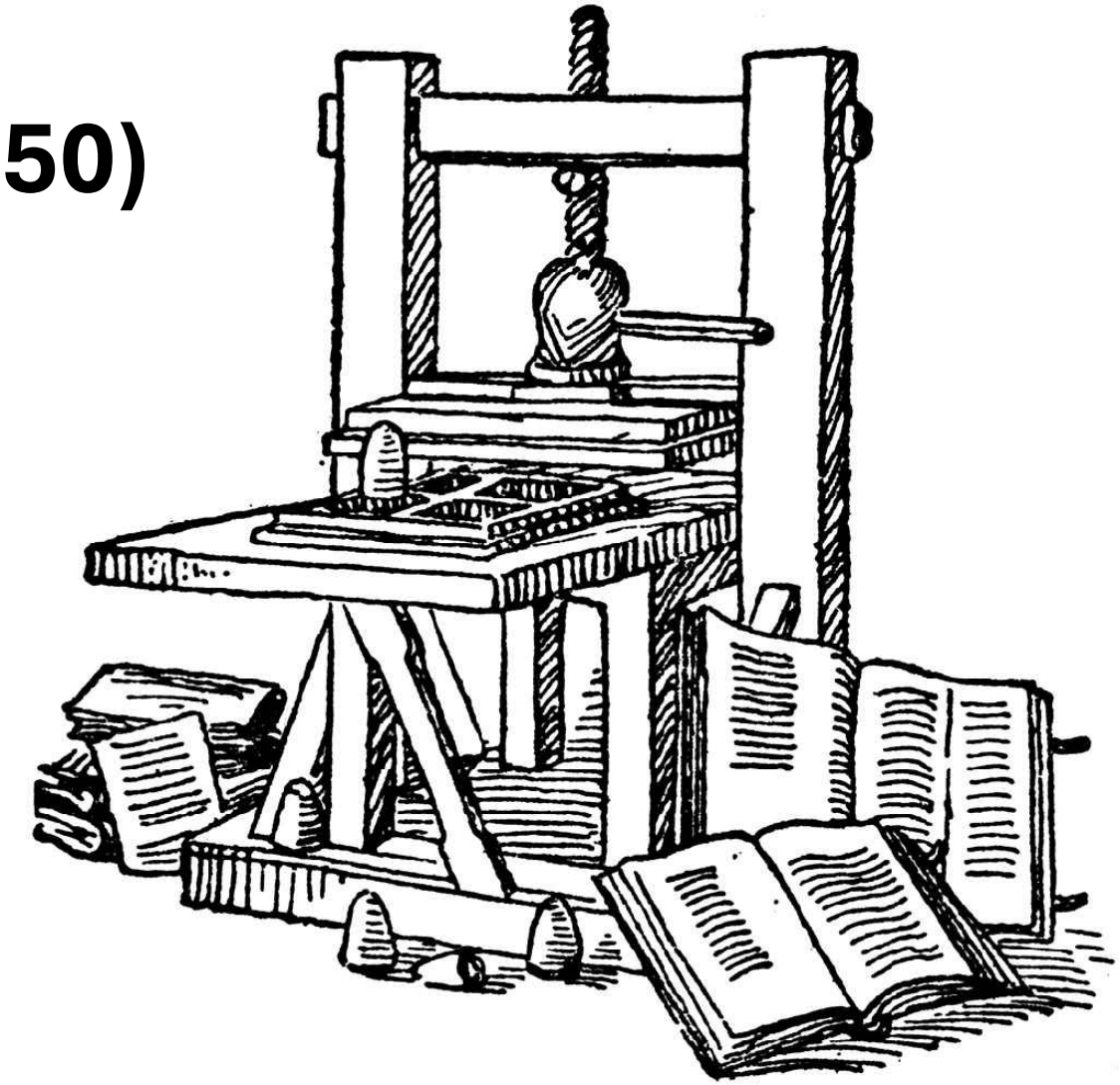






Printing Press Gutenberg (1450)

Gutenberg blended the flexibility of a coin punch with the power of a mechanized wine press to invent mechanized printing.



Creativity is accessible through the often-subconscious process of blending what you already know to form new ideas.

Jeevtronics – Hand-Cranked Defibrillator

Radio
Hand – Cranked



+



Defibrillator

Hand – Cranked Defibrillator





Innovative Idea

In 2000 Inventor and Engineer Ram Dhurjaty was listening to a story about hand crank radios in Africa. He was also quite familiar with battery problems in defibrillators and so he asked the question, Why not have a hand crank defibrillator?

In India Alone 1 Billion of the 1.3 Billion Population does not have easy access to life saving technology like a defibrillator.

A large percentage of tested defibrillators, around 25%, fail to work due to battery charge issues or other electrical problems.

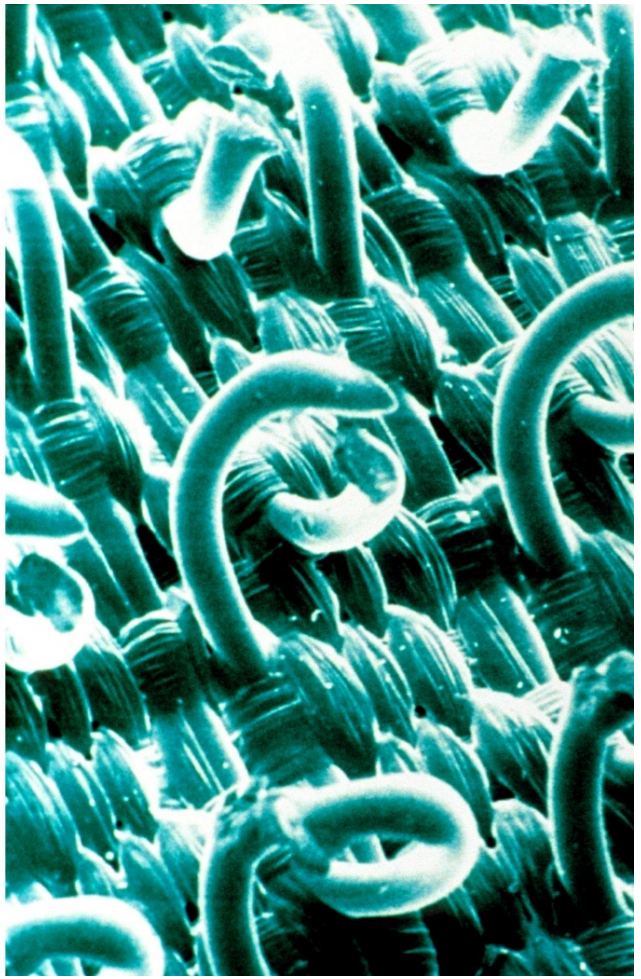
Much of the world is without reliable power. Huge portions of South Asia and Sub-Saharan Africa do not have access to electricity.

C R E A T E



Innovations adapted from Nature - Velcro !





Innovations adapted from Nature – Bullet Train

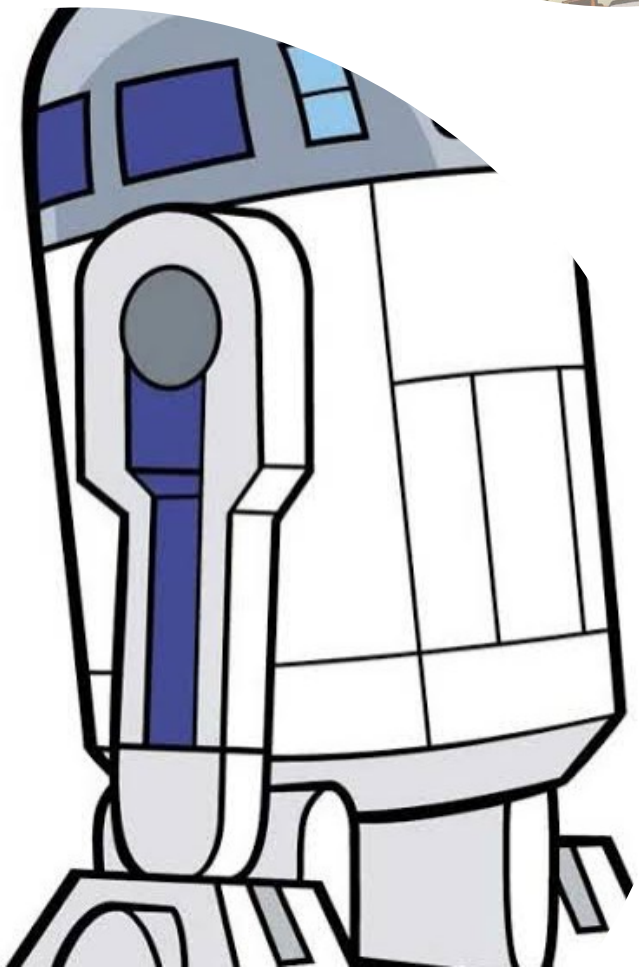


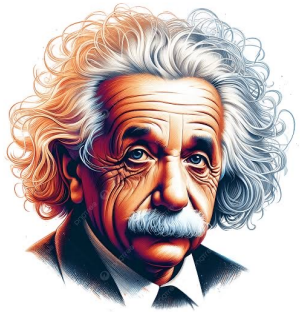










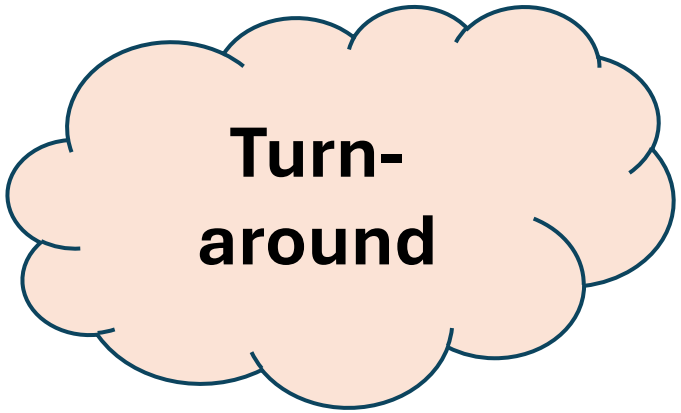


Adapt





C R E A T E





Gatorade

- The drink's origins can be traced to Bangladesh, where for centuries practitioners of ayurvedic medicine have used liquid concoctions of carbohydrates, sugar, and salt to quickly hydrate patients.

- U.S. doctors who traveled to Bangladesh during cholera outbreaks in the early 1960s "discovered" this remedy and wrote about its efficacy.

- And a doctor at the University of Florida read their report and realized that it could also work for dehydrated football players.

C R E A T E



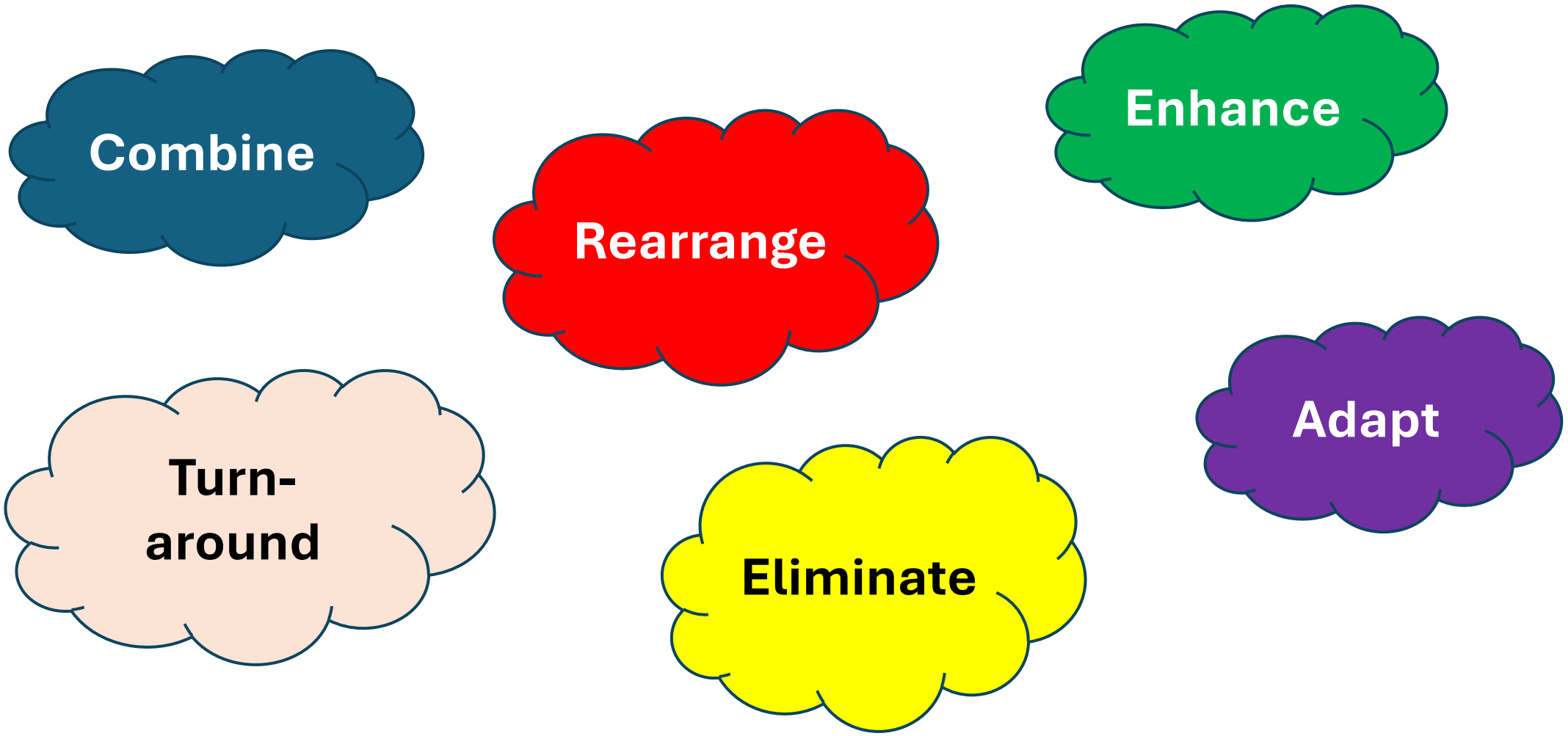

Electric Car

Autonomous Car



Eliminate Steering, Accelerator, Break Pad
Eliminate Driver !!!

CREATE



CREATE - Divergent Thinking Tool

Idea Triggers	How to Improve the Useful Functions - ...	How to Reduce the Harmful Functions -	How to Reduce the Cost of .....
Combine			
Rearrange			
Enhance			
Adapt			
Turn-around			
Eliminate			

!nnovation Flow - Idea Canvas

Ideal Solution :

Useful functions ↑ -

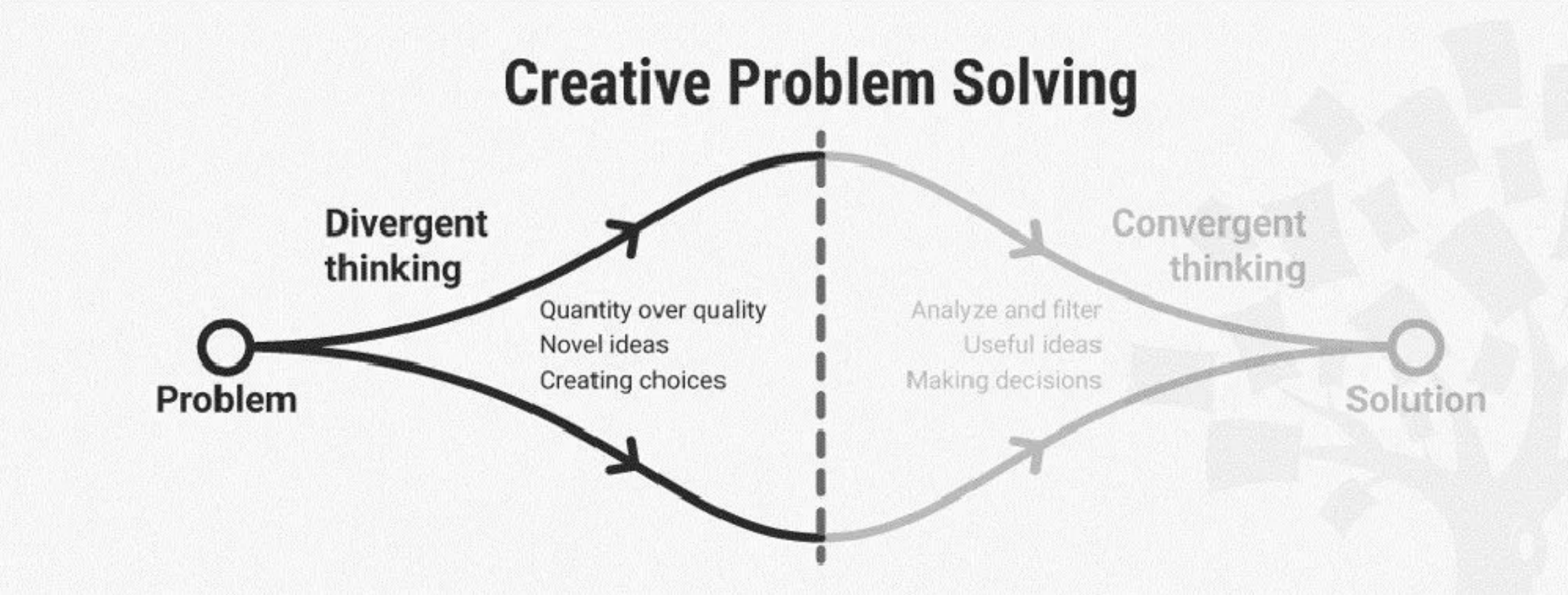
Harmful functions ↓ -

Cost ↓ -

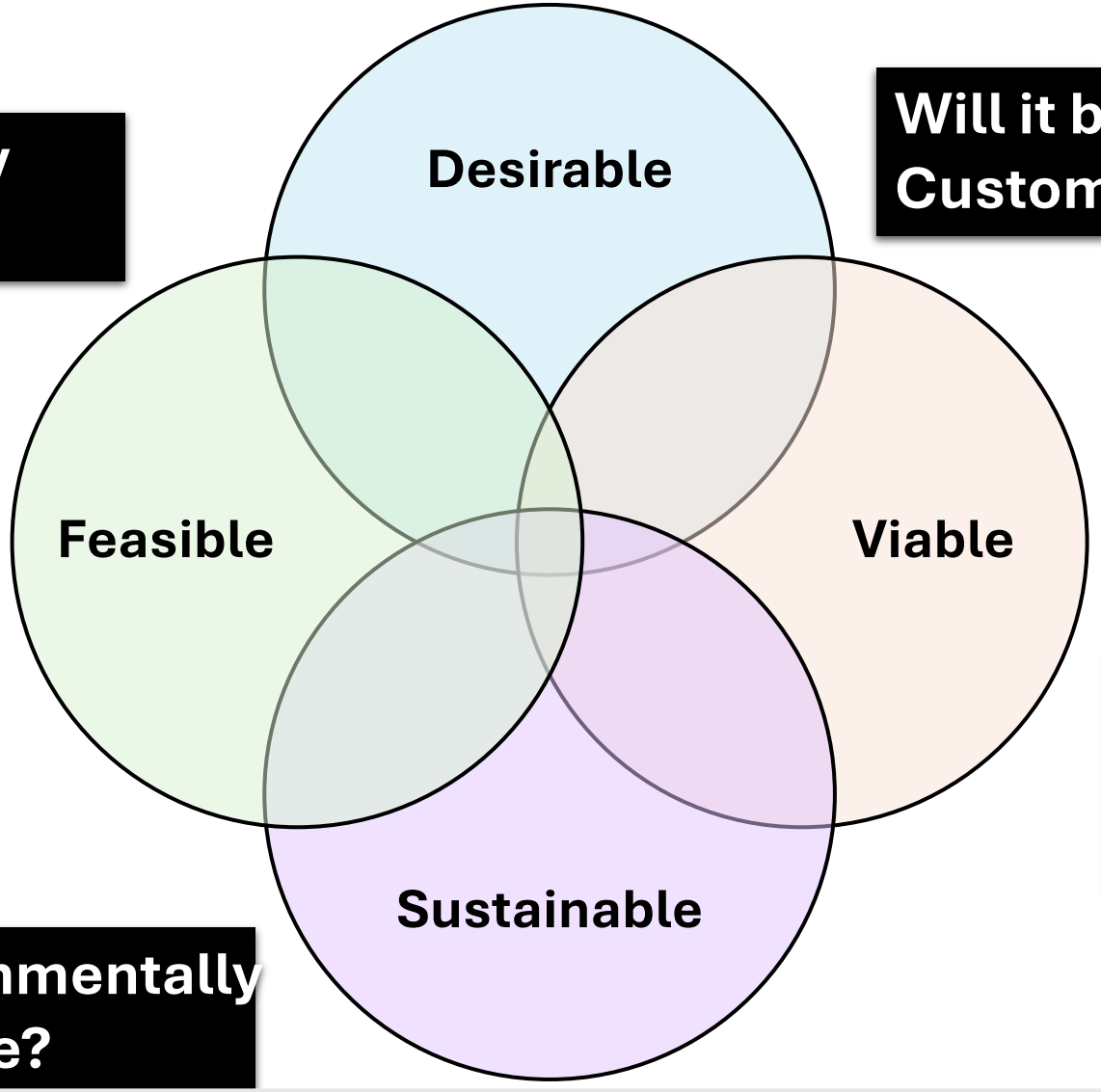
	Useful ↑	Harmful ↓	Cost ↓
Combine			
Rearrange			
Enhance			
Adapt			
Turn - around			
Eliminate			

Innovation Flow – Workshop




Creative Problem Solving
Divergent thinking
Quantity over quality
Novel ideas
Creating choices
Convergent thinking
Analyze and filter
Useful ideas
Making decisions
Problem
Solution

Human Centered Design (HCD) – Idea Filters



Is it technically feasible ?

Will it be desirable to the Customer?

Is it economically viable ?

Is it environmentally sustainable?

!nnovation Flow - Idea Canvas

Ideal Solution :

Useful functions ↑ -

Harmful functions ↓ -

Cost ↓ -

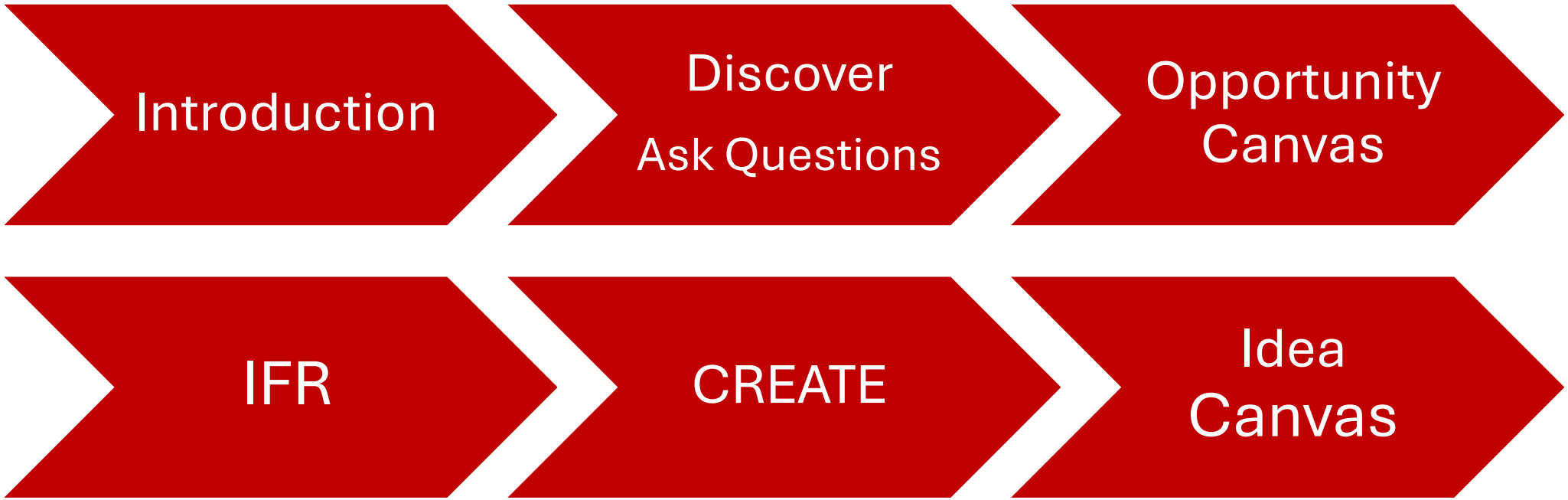
	Useful ↑	Harmful ↓	Cost ↓
Combine			
Rearrange			
Enhance			
Adapt			
Turn - around			
Eliminate			

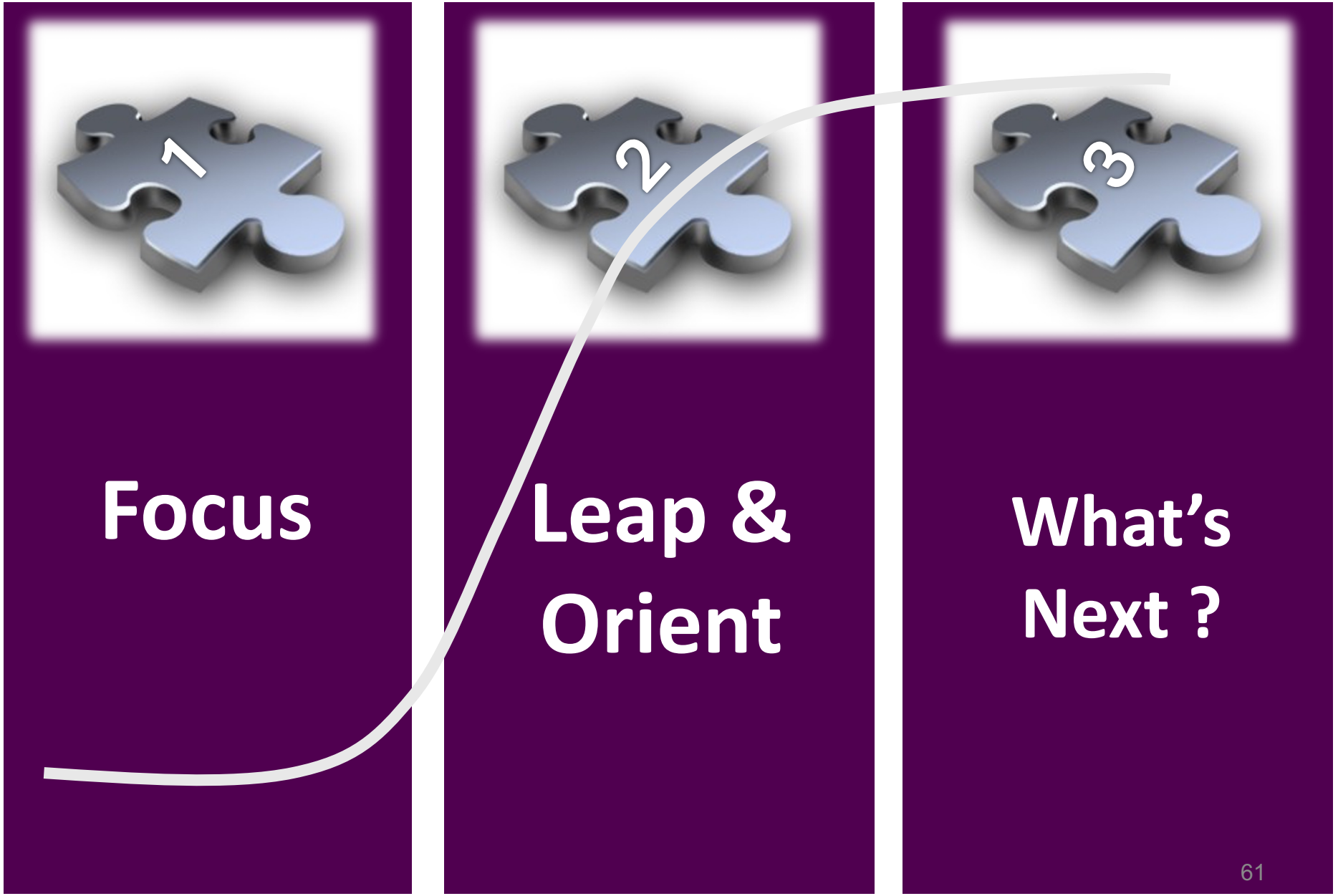
Idea	D	F	V	S

Innovative Thinking

- Innovation FLOW
- Discovery Questions
- Innovation FLOW – Opportunity Canvas
- CREATE
- Innovation FLOW – Idea Canvas

Innovation Flow – Workshop





61

!nnovation Flow – Opportunity Canvas

The Innovation Plot

- **Who is affected ?**
- **What is the problem ?**
- **When is it a problem ?**
- **Where is it a problem ?**
- **How is it a problem ?**
- **Why did the problem arise ?**

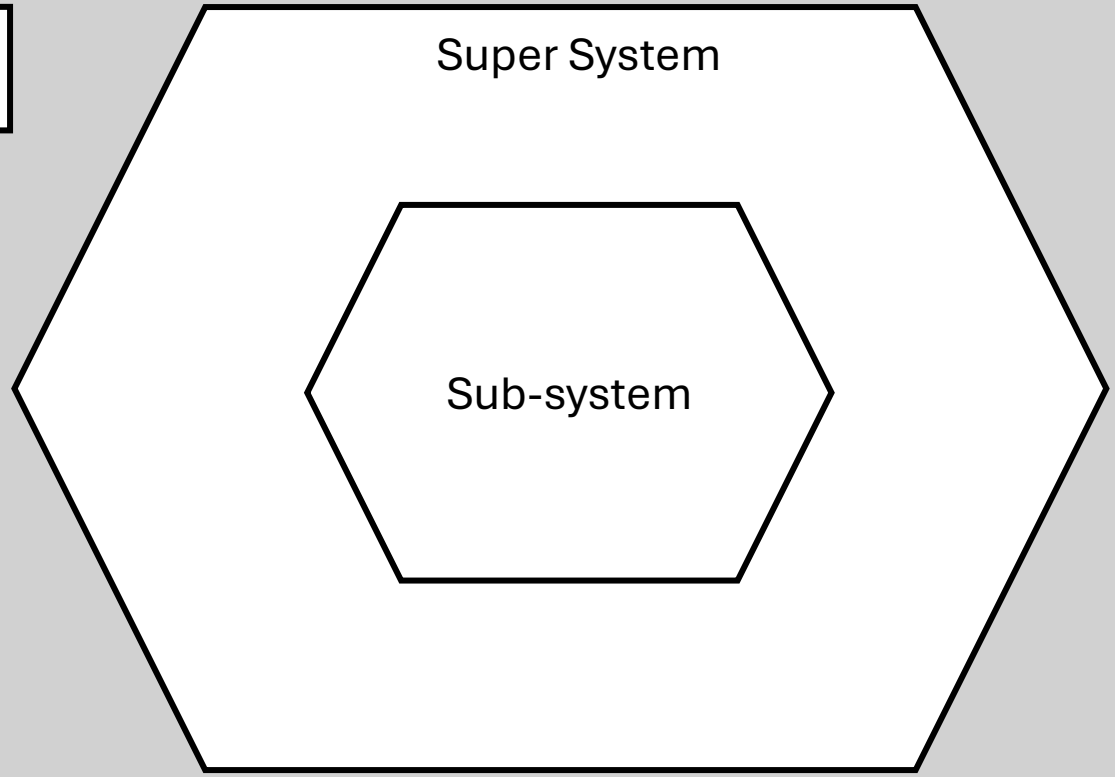
Why

Why

Why

Why

Why



!nnovation Flow - Idea Canvas

Ideal Solution :

Useful functions ↑ -

Harmful functions ↓ -

Cost ↓ -

	Useful ↑	Harmful ↓	Cost ↓
Combine			
Rearrange			
Enhance			
Adapt			
Turn - around			
Eliminate			

Idea	D	F	V	S

!nnovation FLOW

Dr Shankar

IIT Madras

September 2026

